RESILIENCE BY DESIGN

FROM ATOMIC SAFETY TO CLOUD-NATIVE AI

SPRING 2024 MIDTERM MASTERY

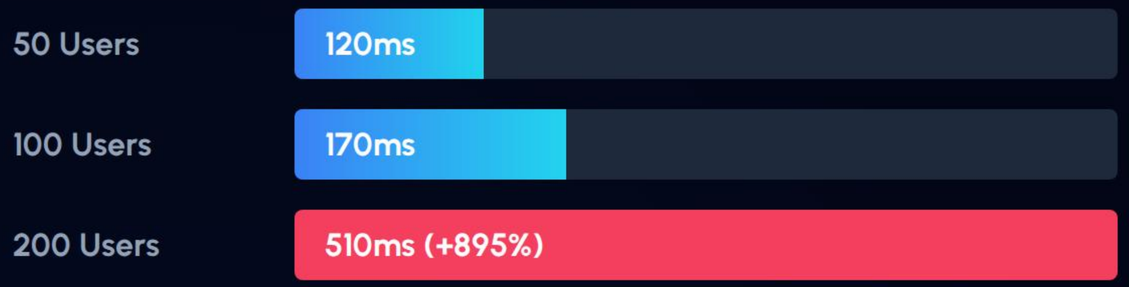
IDENTIFYING THE BREAKING POINT

Load testing analysis from HW6: The transition from stable linear scaling to saturation.

USER COUNT	P50 LATENCY	P99 LATENCY	CPU AVG	HEALTH STATUS
5 Users	36ms	57ms	~0.3%	Healthy
50 Users	38ms	120ms	27.6%	Healthy
100 Users	43ms	170ms	~49.0%	Degrading
200 Users	89ms	510ms	~53.4%	Saturation

Scaling Insight: Identified 27% CPU as the "Perfect Performance" ceiling.

THE PERFORMANCE WALL



AUTO-SCALING TARGET

Decision: Set threshold to **40% CPU**. This ensures scale-out occurs before the 50% saturation cliff.

DISTRIBUTED SYSTEMS CHAIN

CORRECTNESS

Hardware-level atomicity. Correctness is the prerequisite for scaling. If the bottom link fails, the cloud collapses.

EFFICIENCY

Data structures & locks. Trading off Mutex serialisation vs sync.Map for read-heavy distributed workloads.

ELASTICITY

ALB & Auto-Scaling. Horizontal scaling provides the self-healing capability that vertical scaling lacks.

WHEN SCALING IS NOT ENOUGH

Cloud infrastructure cannot fix broken code. Simulated a fatal concurrency race condition in Go.

```
~ $ curl http://localhost:8080/crash
fatal error: concurrent map writes
goroutine 127 [running]:
main.main.func1(0xc0000...)
resilience_demo.go:23 +0x45

curl: (7) Connection Refused
```

BULKHEAD & FAIL FAST

100%

PROCESS UPTIME

The system survived 200 concurrent requests by sacrificing overflow traffic via 503 errors.

```
// Pattern: Resource Isolation
Code: 200 ✅ Handled
Code: 200 ✅ Handled
Code: 503 ❌ Bulkhead Full
Code: 503 ❌ Bulkhead Full
Code: 503 ❌ Bulkhead Full

Status: Healthy (Uptime Preserved)
```

IMAGE INTELLIGENCE PIPELINE

PRODUCER POOL

Decoupled image ingestion using distributed message queues for high-concurrency resizing.

RESILIENT WORKERS

Implementing the Bulkhead pattern to ensure metadata API stability during heavy AI filtering.

DENSITY SCALING

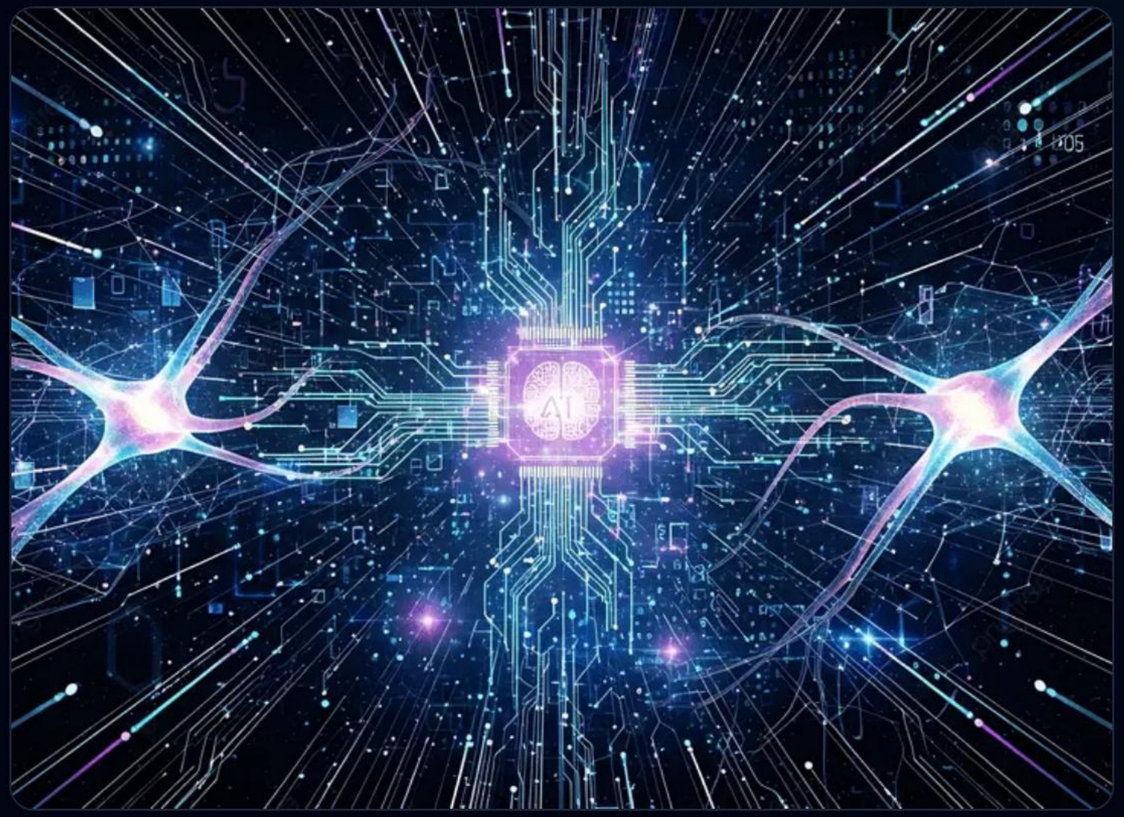
Optimizing workers for **t3.medium** instances by investigating 4-bit model quantization.

INNOVATION: 4-BIT QUANTIZATION

REDUCING MEMORY PRESSURE

Investigating the deployment of quantized ML models to fit more processing threads on constrained nodes.

- 70% reduction in model memory footprint.
- Prevents OOM events in Sandbox environments.
- Maximizes worker density per CPU core.



BUILDING FOR RESILIENCE

"Correctness at the bottom. Efficiency in the middle. Scalability at the top."

PROPOSAL LIVE ON PIAZZA

Yumeng Zeng | Midterm Mastery 2024

IMAGE SOURCES

https://static.vecteezy.com/system/resources/previews/012/935/575/non_2x/abstract-hi-tech-cloud-connection-technology-on-blue-background-cloud-computing-big-data-and-futuristic-business-network-concept-vector.jpg

Source: www.vecteezy.com

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https://static.vecteezy.com/system/resources/previews/068/068/488/non_2x/increasing-trend-with-red-growth-chart-showing-progress-and-positive-performance-in-financial-metrics-png.png

Source: www.vecteezy.com

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Source: www.istockphoto.com

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Source: pngtree.com